

Three piles break the mirror

Name: _____

Date: _____

Build three piles. On a turn, remove any positive number of counters from exactly ONE pile. Take the last counter to win. Use ordinary counters, not bundles, for actual play.

1. Challenge an adult.

Try piles $(1, 2, 3)$, then $(1, 2, 4)$. Change who moves first. Which starts seem to be traps for the player receiving them?

Record moves, replies, and a conjecture.

2. Compare two kinds of balance.

With only two piles, matching sizes lets you copy your opponent. With three piles, investigate $(1, 1, 1)$, $(1, 1, 2)$, $(1, 2, 3)$, and $(2, 4, 6)$. Is an even *total* enough to make a trap?

Find evidence against a rule that almost works.

3. Trade for bundles.

Use labels 1, 2, and 4. Write each pile as a sum using each label at most once: for example, 5 is $4+1$. For each label, count how many piles use it. Which starts have an even count for every bundle?

Try before turning the page. An adult can make the bundle labels with scraps of paper.

Try it with objects. Explain by pointing, talking, or drawing.

Balance the binary columns

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Represent each pile using bundles 4, 2, 1. Write 1 if that bundle is used and 0 if not. Call a position **balanced** when every column contains an even number of 1s. Zero 1s counts as even.

Pile	4	2	1	Pile	4	2	1
1	0	0	1	1	0	0	1
2	0	1	0	2	0	1	0
3	0	1	1	4	1	0	0

4. Repair the unbalanced position.

Change just one pile in (1, 2, 4) to a *smaller* number, making every column even. What move does this mean with counters?

5. Test your rule.

For each start, either give a move to balance or say it is already balanced. Then play against someone trying to break your prediction.

Start	4/2/1 records	Move or already balanced?
(2,4,6)		
(3,4,5)		
(1,5,7)		
(3,5,6)		

Can a legal move from balance leave every column even? Why?

Vocabulary after discovery: The column parities form the *Nim sum*, also called bitwise XOR. A balanced position has Nim sum zero.

Try it with objects. Explain by pointing, talking, or drawing.

Why must the strategy work?

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6. The highest wrong column.

Suppose some column is odd. Find the largest bundle whose column is odd. At least one pile uses it. In that pile:

1. Keep all larger bundle choices unchanged.
2. Remove this bundle.
3. Choose the smaller bundles to make *every* column even.

Try this recipe on $(3, 4, 5)$. Then try $(7, 10, 12)$, adding an 8-column.

My moves, and why they fix all the columns:

7. Two missing pieces of a proof.

Why is the new pile *smaller*, even if you add some smaller bundles? Why can the game not continue forever?

A reason that works for any pile sizes:

8. Design a trap.

Choose any first two piles. Can you choose the third to make balance? Could you do it in two different ways? Give your puzzle and its explanation to an adult.

My designed start and why its third pile is forced:

Try it with objects. Explain by pointing, talking, or drawing.